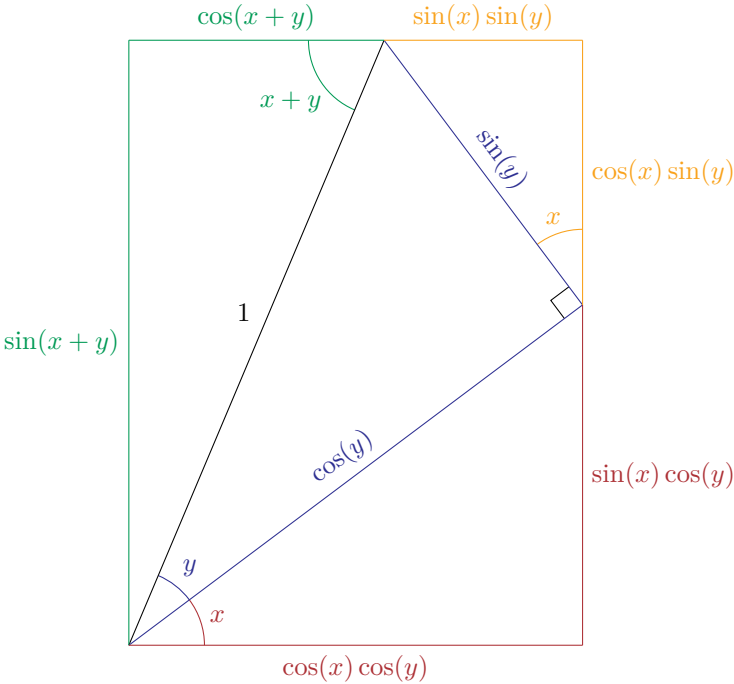
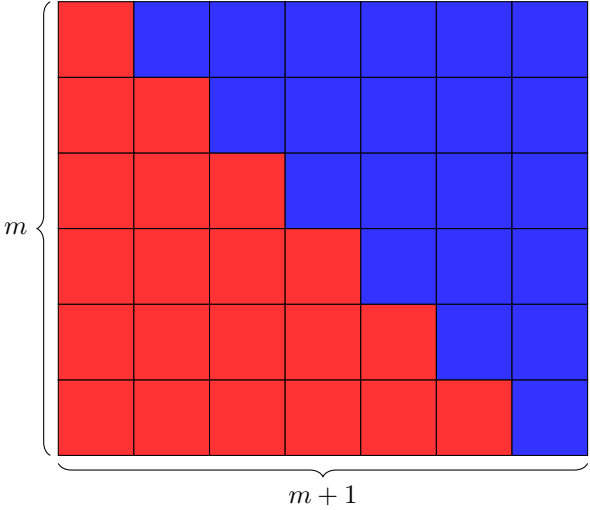


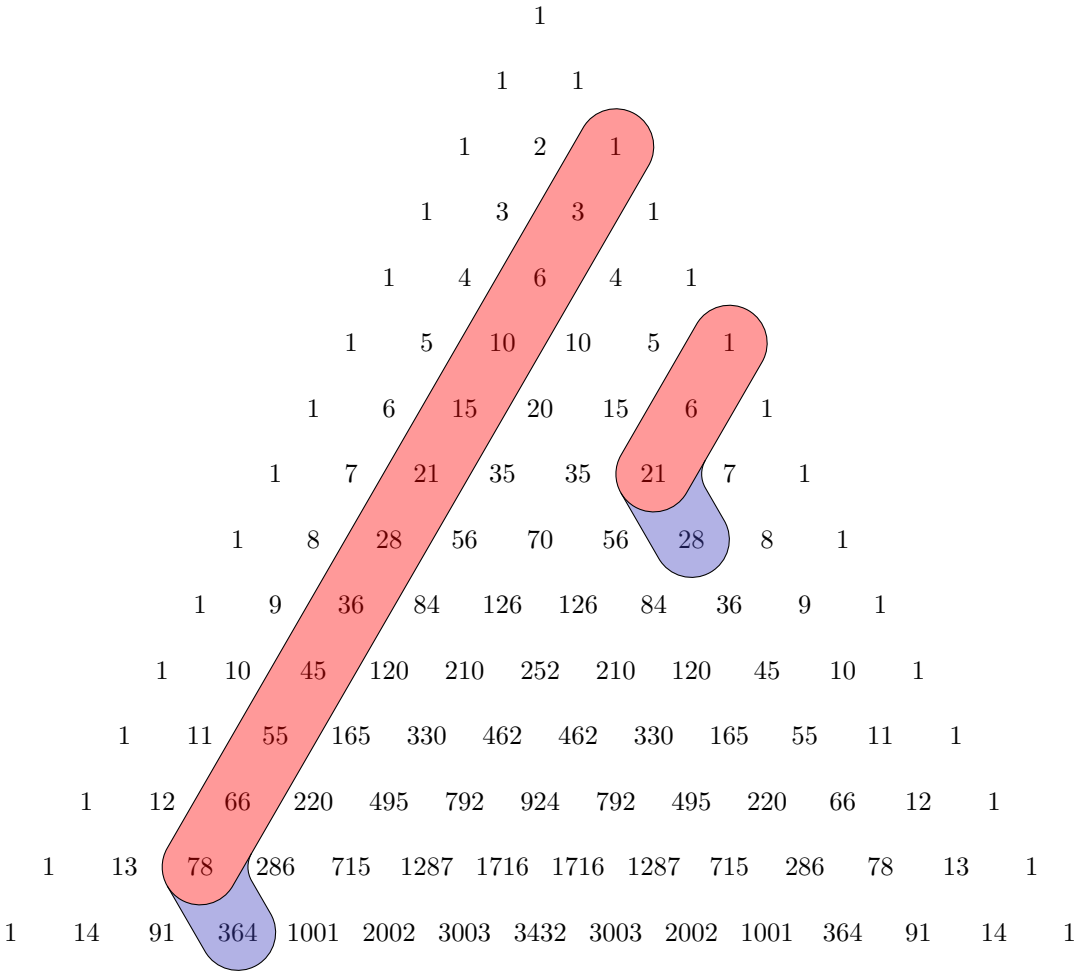


$$\sin(x + y) = \sin(x) \cos(y) + \cos(x) \sin(y)$$
$$\cos(x + y) = \cos(x) \cos(y) - \sin(x) \sin(y)$$

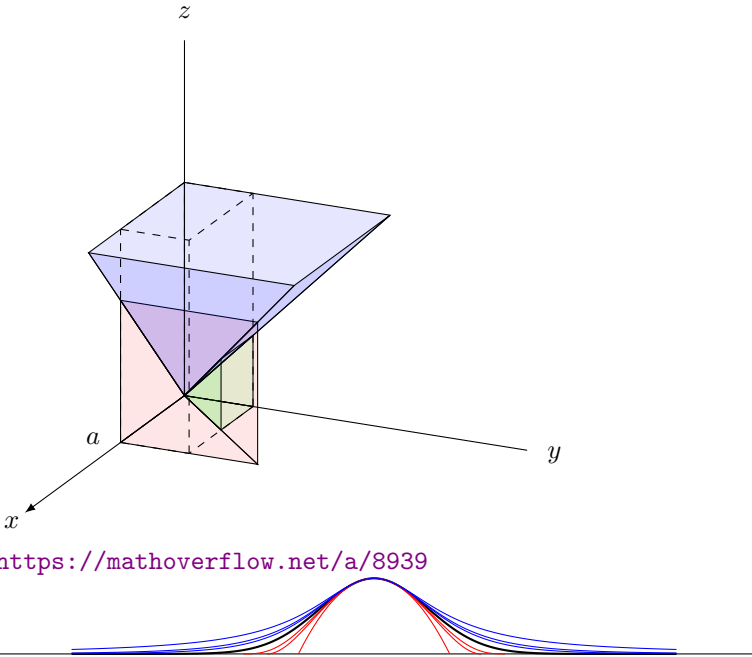


Theorem -1.1 (Hockey Stick Identity). For all $n, k \in \mathbb{N}_0$, we have:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$$



$$\frac{a^3 + b^3 + c^3}{3} \geq abc$$



$$\begin{aligned}
 \int_{-\sqrt{n}}^{\sqrt{n}} \left(1 - \frac{x^2}{n}\right)^n dx &= \sqrt{n} \int_{-1}^1 (1 - y^2)^n dy && (x = \sqrt{n}y) \\
 &= \sqrt{n} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t)^{2n+1} dt && (y = \sin(t))
 \end{aligned}$$